

BASE BERNARDO O'HIGGINS, Antarctica

WMO: 890590

Lat: 63.321S

Lon: 57.899W

Elev: 33

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.1	1.0	-7.6	3.7	-0.3	-4.1	4.4	3.1	92.0	7.9	72.4	10.6	35.8	150	1.081

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.4	38.5	35.1	37.3	34.7	36.3	34.2	36.4	37.3	35.6	36.4	34.9	35.7	14.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
35.6	30.6	36.4	34.8	29.6	35.7	34.0	28.6	35.0	13.8	37.5	13.4	36.6	13.1	35.9	45.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
63.3	53.3	45.4	-7.4	45.1	5.4	5.3	-11.3	48.8	-14.5	51.9	-17.5	54.9	-21.4	58.7		
			-7.1	39.4	5.2	2.2	-10.8	41.0	-13.9	42.2	-16.8	43.5	-20.6	45.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	33.5	32.7	30.1	26.5	24.3	20.1	17.8	19.4	21.7	25.2	28.8	32.0
	DBStd	7.73	1.94	2.58	4.00	5.69	5.99	7.46	8.33	7.53	7.78	5.57	3.55	2.45
	HDD50	8766	510	484	617	704	798	898	997	947	848	769	635	559
	HDD65	14241	975	904	1082	1154	1263	1348	1462	1412	1298	1234	1085	1024
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.1	12.1	12.2	14.5	15.3	14.8	17.4	16.4	16.8	16.6	16.4	14.6	13.8
Precipitation	PrecAvg	36.2	2.7	2.9	3.8	3.2	3.0	2.4	2.6	2.7	3.1	3.8	3.3	2.6
	PrecMax	49.2	4.2	4.5	5.4	5.9	4.7	4.1	5.8	5.0	4.4	5.2	5.0	4.2
	PrecMin	28.5	1.8	2.1	2.4	1.8	1.6	1.1	0.9	0.7	1.4	2.5	1.6	1.1
	PrecStd	4.8	0.6	0.6	0.7	0.9	0.7	0.7	1.0	1.1	0.7	0.6	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	39.7	38.1	36.0	35.4	33.6	33.2	33.1	34.1	34.2	37.3	39.6
		MCWB	37.2	36.2	35.1	34.4	33.0	32.0	31.4	31.8	32.4	32.1	34.5	35.5
	2%	DB	38.5	37.6	36.6	34.8	33.6	32.0	31.8	31.9	32.6	33.1	34.8	37.4
		MCWB	35.1	35.4	34.6	33.5	32.5	31.1	30.7	30.9	31.3	31.8	32.8	34.2
	5%	DB	37.3	36.7	35.6	33.9	32.5	31.0	30.8	30.9	31.8	32.3	33.8	35.9
		MCWB	34.5	34.9	34.2	32.8	31.5	30.0	29.8	30.1	30.8	31.2	32.1	33.4
	10%	DB	36.3	35.9	34.8	33.0	31.6	29.9	29.5	29.9	31.0	31.6	33.0	34.9
		MCWB	33.9	34.3	33.5	32.0	30.7	28.9	28.6	29.2	30.1	30.6	31.5	32.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	37.6	36.6	35.3	34.1	32.3	32.2	32.3	32.7	33.1	35.0	36.6
		MCDB	39.6	38.3	37.1	35.6	34.7	32.7	32.6	32.6	33.4	33.4	36.0	38.5
	2%	WB	36.3	36.3	35.5	34.0	32.9	31.4	31.2	31.2	31.7	32.2	33.6	35.0
		MCDB	37.4	36.9	35.9	34.4	33.3	31.8	31.6	31.6	32.2	32.6	34.4	36.4
	5%	WB	35.3	35.5	34.7	33.1	31.9	30.4	30.2	30.3	31.1	31.5	32.7	34.0
		MCDB	36.3	36.1	35.2	33.5	32.3	30.9	30.7	30.8	31.6	32.0	33.4	35.3
	10%	WB	34.5	34.8	33.8	32.2	30.9	29.2	28.8	29.3	30.3	30.9	31.9	33.2
		MCDB	35.7	35.5	34.5	32.8	31.5	29.8	29.5	29.9	30.9	31.5	32.7	34.4
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.4	5.3	5.8	6.8	7.8	8.7	8.6	7.3	5.9	4.8	4.4
		MCDBR	6.2	5.6	5.5	5.7	6.1	7.2	9.1	8.8	5.5	5.3	5.4	5.9
		MCWBR	4.8	4.5	4.9	5.3	5.8	7.0	8.7	9.0	5.4	5.2	4.9	4.8
	5% WB	MCDBR	5.4	4.8	4.9	5.1	5.6	6.3	7.6	8.4	5.4	5.1	5.1	5.4
		MCWBR	4.8	4.4	4.8	5.1	5.6	6.6	7.9	8.8	5.6	5.2	4.9	4.7
Clear-Sky Solar Irradiance	taub	0.293	0.290	0.275	0.249	0.184	0.157	0.172	0.241	0.266	0.288	0.288	0.293	
	taud	2.383	2.398	2.410	2.331	2.029	1.919	1.987	2.205	2.293	2.306	2.340	2.351	
	Ebn at Noon	310	295	270	220	143	81	142	214	266	291	310	314	
	Edh at Noon	34	30	23	16	9	4	8	17	26	32	36	36	
All-Sky Solar Radiation	RadAvg	1758	1238	674	287	81	26	54	221	617	1185	1743	1990	
	RadStd	112	69	50	20	5	1	2	15	39	64	99	139	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page